A3-PS013-TC046

Incident Timeline & Evidence Collection for Small Teams

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AIORI Hackathon 3

October 10, 2026

Problem Description

The Solo Defender's Forensic Bottleneck

- **Clock Drift Chaos:** Unaligned local clocks and inconsistent time zones obscure critical attack causality.
- **Fragile Evidence Integrity:** Manual analysis risks accidental file alteration without cryptographic chain of custody.
- **Syntax Fragmentation:** Linux sys logs and webserver access logs feature fundamentally mismatched syntaxes.
- **High Triage Latency:** Ad-hoc manual spreadsheet correlation delays critical threat containment decisions.

Background

Core Principles of Incident Telemetry

- **Evidence Immutability:** Digital evidence requires strict segregation between read-only source logs and normalized analytical copies to uphold evidentiary integrity.
- **Temporal Normalization:** Distributed systems run diverse timezone offsets and NTP skews. Reconstructing genuine sequence order demands uniform UTC transformation.
- **Schema Normalization:** Disparate unstructured strings must map into standard fields: Actor, Action, Entity, Status, and Timestamp without data truncation.

Literature & RFC Review

Applicable Standards & RFC Specifications

- **NIST SP 800-86:** Guide to Integrating Forensic Techniques into Incident Response → Defines the standard 4-step forensic lifecycle: Collection, Examination, Analysis, and Reporting with strict integrity controls.

- **ISO/IEC 27037:** Guidelines for Identification, Collection, Acquisition & Preservation→Specifies non-repudiation benchmarks and cryptographic integrity verification standards for raw digital evidence.

- **RFC 5424 / RFC 3339:** The Syslog Protocol & Date/Time Specification on the Internet. Governs syslog grammar, high-precision timestamp headers, and universal timezone offset parsing.

Proposed Solution

Lightweight Forensic Pipeline Architecture

1. **Immutable Ingest:** Automatic raw record preservation with instant SHA-256 manifest calculation on ingress.

2. **Dual Parser Engine:** High-throughput parsers supporting Linux Syslog/Auth and Web Server (Nginx/Apache) logs.

3. **UTC & Skew Resolver:** Per-source clock delta configuration aligning multi-host timestamps to a single synchronized timeline.

4. **Timeline & Export:** Fast chronological search, filtering by IP/user/severity, and exportable forensic executive reports.

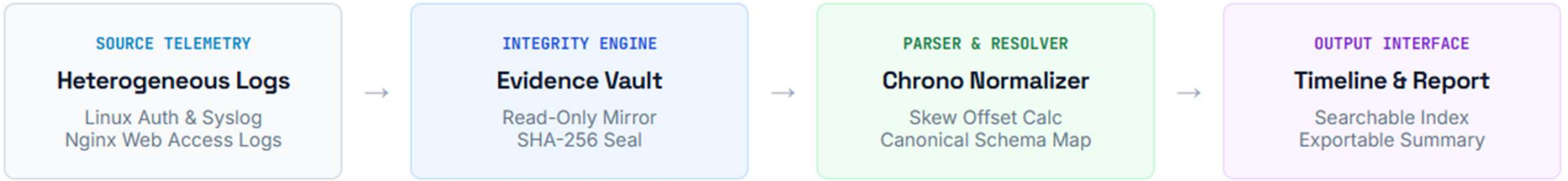
Optimization Proposed

Evaluation Metric	Manual Baseline (Greps / Sheets)	PS-013 Optimized Tool
Triage Latency	45 to 90 minutes per incident	Under 3 minutes automated
Timestamp Synchronization	Manual offset calculation & prone to error	Automated UTC offset engine
Evidence Verification	No verification; files modified in place	Instant SHA-256 evidence sealing
Chronological Correlation	Manual copy-paste spreadsheet merging	Unified chronological event graph
Unparsed Log Handling	Silent omission during ad-hoc filters	Explicit parsing gap ledger & audit

AIORI 3 Architecture Design (UML Diagram)

Solution Architecture & Design

Modular system design enforcing passive forensic integrity and read-only boundaries



Cope Boundary Check: Strictly passive log processing — zero endpoint modification, malware termination, or live process tampering.

Data Flow Diagram

End-to-end data pipelines and state transitions across the 3 synthetic incident scenarios

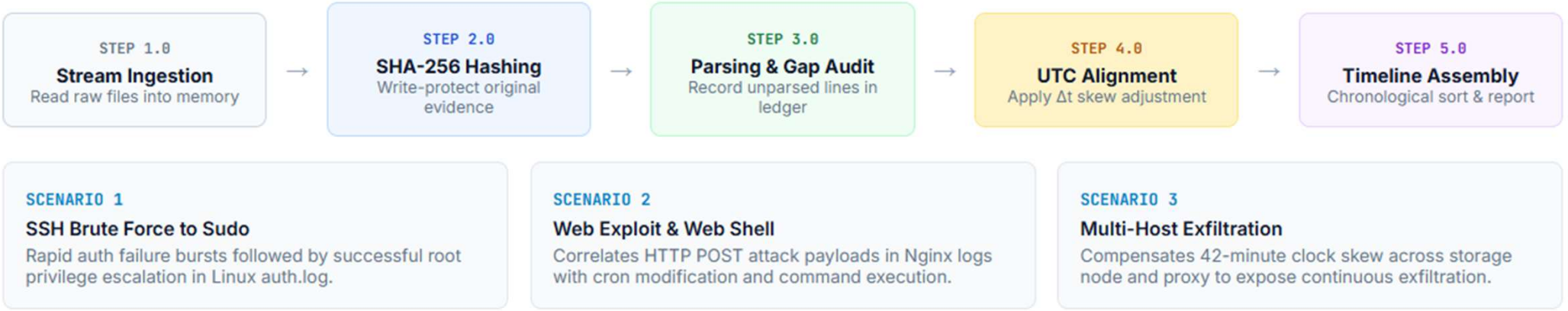


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Hackathon Milestones & Methodology

1. **Repo Setup & Ingestion:** Private GitHub repository setup with @aiori-hackathon collaborator onboarding; SHA-256 evidence hashing module.

2. **Parsers & Skew Engine:** Implementation of dual syslog & web access log parsers, UTC converter, and explicit parsing-gap audit logger.

3. **Timeline UI & Scenarios:** Execution of 3 synthetic incident scenarios; chronological correlation engine with query and filter controls.

4. **Demonstration & PDF:** Prototype demonstration with pseudocode walkthrough to mentor; Proposed-structure-hackathon.pdf upload.

Bibliography I

1 **[1] Kent, K., Chevalier, S., Grance, T., & Dang, H. (2006).** *Guide to Integrating Forensic Techniques into Incident Response*. NIST Special Publication 800-86.

2 **[2] International Organization for Standardization. (2012):***Information technology — Security techniques — Guidelines for identification, collection, acquisition and preservation of digital evidence*. ISO/IEC 27037:2012.

3 **[3] Gerhards, R. (2009).:** *The Syslog Protocol*. RFC 5424, Internet Engineering Task Force (IETF).

4 **[4] Klyne, G., & Newman, C. (2002).:** *Date and Time on the Internet: Timestamps*. RFC 3339, Internet Engineering Task Force (IETF).

Thank You for Your Attention!